Dr. BIBHUTIBHUSAN NAYAK

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PROFILE:

Highly self-motivated and results-driven researcher with proven expertise in ***magnetic thinfilm based sensor, application of sensor in bio detection and robotics , localization , autonomous driving, advanced materials characterization****, and* ***magneto-functional sensor development****. Demonstrates strong analytical thinking, problem-solving abilities, and effective communication skills in collaborative research environments.*

RESEARCH INTERESTS:

- Geomagnetic field mapping and localization for autonomous robots operating in GPS-denied and visually challenging environments.
- Multi-modal sensor fusion using magnetic sensors, IMUs, cameras, LiDAR, and GNSS for robust navigation and state estimation.
- Learning-based magnetic field modeling and neural magnetic maps for localization, place recognition, and SLAM applications.
- Development of magnetic sensing systems and algorithms for long-term autonomous navigation in indoor, underground, and urban environments.
- Physics-guided simulation and modeling of magnetic fields and sensor responses to support localization, mapping, and robotic decision-making.

RESEARCH AND TECHNICAL EXPERIENCE:

❖ Robotics Localization & Navigation

- Developed localization and navigation systems for autonomous robots operating in GPS-denied and vision-degraded environments.
- Designed and implemented geomagnetic field mapping, place recognition, and magnetic-aided localization frameworks for indoor and outdoor navigation.
- Worked on multi-modal localization using magnetometers, IMUs, cameras, LiDAR, GNSS, and barometric sensors.
- Evaluated localization performance using benchmark datasets and real-world robotic deployments.

❖ Sensor Fusion & State Estimation

- Developed sensor fusion pipelines integrating magnetic sensors, IMUs, cameras, LiDAR, GNSS, and pressure sensors.
- Implemented state-estimation algorithms including Kalman Filters, Extended Kalman Filters (EKF), complementary filters, and optimization-based approaches.
- Performed sensor calibration, synchronization, coordinate transformation, and uncertainty analysis for multi-sensor robotic systems.
- Designed real-time data processing frameworks for robust localization and navigation.

❖ ROS/ROS2 & Autonomous Systems

- Extensive experience with ROS and ROS2 for robotic perception, localization, mapping, and autonomous navigation.

- Developed custom ROS2 nodes for real-time acquisition and synchronization of multi-modal sensor data.
- Experience with TF transformations, coordinate frames (SO(3)/SE(3)), rosbag processing, sensor drivers, and robotic middleware.
- Integrated LiDAR, cameras, IMUs, magnetometers, and GNSS sensors on robotic platforms.

❖ Ground Truth Generation & Dataset Development

- Developed large-scale robotic datasets containing synchronized LiDAR, camera, IMU, magnetometer, and GNSS measurements.
- Generated ground-truth trajectories using LiDAR-based mapping, pose-graph optimization, and multi-sensor calibration frameworks.
- Designed benchmarking protocols and evaluation metrics for localization and navigation systems.
- Experience in data collection, annotation, synchronization, and validation for robotic perception research.

❖ Data Acquisition & Embedded Systems

- Designed and implemented real-time sensor acquisition systems for robotics and autonomous platforms.
- Developed sensor interfacing, signal conditioning, embedded communication, and wireless telemetry systems.
- Experience with serial communication, SPI, I2C, USB, Ethernet, and real-time sensor integration.
- Built automated pipelines for sensor logging, monitoring, and data management.

❖ Machine Learning & Scientific Computing

- Strong programming skills in Python, C/C++, MATLAB, SQL, and Linux environments.
- Experience with TensorFlow, Keras, PyTorch, and machine learning for localization, mapping, and sensor data analysis.
- Developed automated pipelines for data processing, visualization, filtering, and AI-driven sensor interpretation.
- Experience with large-scale dataset analysis, neural field modeling, and data-driven environmental representation.

❖ Simulation & Digital Twins

- Experience with simulation and modeling of robotic sensing systems using MATLAB, Python, COMSOL, MuMax3, and OOMMF.
- Developed simulation frameworks for sensor evaluation, localization testing, and algorithm validation.
- Applied physics-based and data-driven models to analyze sensing performance in complex environments.
- Created virtual environments and experimental validation pipelines for autonomous robotic systems.

ADDITIONAL EXPERIENCE:

- **Project Management**: Experience in leading multi-disciplinary research projects involving sensor development, data acquisition systems, and machine learning based localization.
- **Collaboration & Communication**: Effective communicator in collaborative research settings, with international research exposure (e.g., bilateral projects, Proposal writing).

EDUCATION:

Degree / Position	Institution / Details	Year
Postdoctoral Researcher	DGIST, South Korea, with **Prof. Giseop Kim**	Sept 2023 – Present
Ph.D. in Physics	IIT Hyderabad, Telangana, India	June 2022
M.Sc. in Physics (First Class)	Ravenshaw University, Odisha, India	June 2017
B.Sc. in Physics (First Class)	Fakir Mohan University, Odisha, India	May 2015

MAJOR COURSES:

- Solid state Physics (Magnetism)
- Quantum Mechanics
- Experimental Techniques
- Computational Physics
- Electronics

ACTIVITIES:

- IEEE RAS member , 2025- present
- Optical Society of America (OSA), Student Member, 2021-2023.
- Magnetic Society of India (MSI) , member , 2019- present
- Korean magnetic society (KMS), member, 2023-presnt

WORK STYLE:

- Willing to perform basic tasks and move on to solve complex problems
- Able to learn new knowledge and adapt to new environments quickly
- Strong independent work style and excellent teamwork skills
- Well-organized and passionate

PUBLICATIONS:

- Development of a Three-Axis Planar Hall Magnetoresistance Sensor Using a Superparamagnetic Nanoparticle-Based Flux Guide. C. Jeon, M. Kim, K. Kim, **Bibhutibhusan Nayak**, *et al.* " Advanced Functional Materials (2026): e76501.[https://doi.org/10.1002/adfm.76501]
- Shape Anisotropy-Induced Local Incoherent Magnetization: Implications for Magnetic Sensor Tuning.

  Jinwoo Kim, **BB Nayak**, Ivan Soldatov, Rudolf Schäfer, CheolGi Kim.(2025).*Journal of Science: Advanced Materials and Devices,10(2),100893*
- Remote detection of bovine serum albumin (BSA) using cantilever beam magnetometer. , **B Nayak** and S. Narayana Jammalamadaka, *Journal of Magnetism and Magnetic Materials* 589 (2024): 171537
- Effect of Low Substrate Temperature on the Magnetic Properties and Domain Structure of $Fe_{70}Ga_{30}$ Thin Films.**B. B. Nayak**, U. M. Kannan, and S. Narayana Jammalamadaka *IEEE Transactions on Magnetics* 56, 1-9(2020).
- Effect of sputtering power on the first order magnetization reversal, reversible and irreversible process in Fe71Ga29 thin films. **B. B. Nayak**, and S. Narayana Jammalamadaka *Journal of Magnetism and Magnetic Materials* 536, 168107(2021).

- Magnetic properties and domain imaging of Fe70Ga30 films. **B. B. Nayak**, UM Kannan, M. M.Raja, &S.N. Jammalamadaka In *AIP Conference Proceedings* (Vol. 2265), No. 1, p. 030555, 2020. AIP Publishing LLC.
- Pseudo magnetic properties and evidence for vortex state in Fe2NiGe Heusler alloy thin films., R. K. Roul,**B .B Nayak**, A.K Jana, S. N. Jammalamadaka, "*Journal of Magnetism and Magnetic Materials* 556,169401 (2022).
- Effect of thickness of FeGa layer on the uniaxial anisotropy, exchange bias and domain structure of IrMn(15nm)/FeGa(t) thinfilms, **B. B. Nayak**, and S. Narayana Jammalamadaka (under preparation)
- Temperature evolution of pseudo magnetic properties and vortex state in Fe71Ga29 thin films. Kumar, P., **B.B. Nayak**, Roul, R. K., & Jammalamadaka, S. N. (2023). *Journal of Magnetism and Magnetic Materials*, *585*, 171155.

PATENT:

- "A SYSTEM AND METHOD OF DETECTION OF BOVINE SERUM ALBUMIN (BSA) USING CANTILEVER BEAM MAGNETOMETER ", INDIA, *Patent No*- 552856, 22/10/2024

SOCIAL PROFILE:

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Google Scholar: https://scholar.google.com/Dr bibhutibhusan Nayak